

Wren Oswin

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Software Engineer • MTech Candidate at IIT Kanpur

Education

Indian Institute of Technology, Kanpur

Kanpur, Uttar Pradesh

Master of Technology in Computer Science and Engineering (GATE CSE: AIR 160, Score: 859) July 2026 – Present

Sree Chitra Thirunal College of Engineering

Thiruvananthapuram, Kerala

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.31/10

Aug. 2019 – July 2023

Experience

Software Engineer

July 2023 – January 2025

Tata Elxsi Ltd.

Thiruvananthapuram, Kerala

- Optimized production OTT app responsiveness by delivering **8–10 UI features** using **JavaScript, LightningJS, and MongoDB**.
- Boosted codebase test coverage from **50% to 80%** by authoring automated unit tests for **5–10 core modules**.
- Resolved **10–15 critical sprint bugs** using log analysis and graphical performance debugging tools.
- Engineered a **confidential playback POC** using **React Native and Shaka Player** to port an OTT application to a new OS.

Software Engineer Intern

Sep. 2022 – June 2023

Tata Elxsi Ltd.

Thiruvananthapuram, Kerala

- Built automated CI/CD infrastructure using **DevOps tooling** (Terraform, Ansible, Docker, Concourse) and core **AWS workflows** (EC2, S3).

Projects

High-Performance Black-Scholes Derivative Pricing Engine

June 2026 – July 2026

Technologies: C++, OpenMP, SIMD Vectorization, Linux, GCC Compiler Tools, Gprof

- Architected an ultra-low latency engine to price 1M+ European options via the analytical Black-Scholes model.
- Accelerated grid volatility calculations by multi-threading intensive pricing loops across multi-core CPUs via **OpenMP**.
- Leveraged **SIMD compiler directives** to enforce hardware-level loop unrolling and parallel register packing.
- Optimized runtime execution using **-O3 and -march=native** compiler flags to minimize cache misses and latency jitter.

SmartInbox: Intelligent Message Classification Engine

March 2026 – April 2026

Technologies: Python, SciKit-Learn, Pandas, Streamlit, TF-IDF

- Built an end-to-end text routing engine classifying incoming channel messages automatically with **96%+ accuracy**.
- Extracted high-value semantic features from unstructured natural text using a custom **TF-IDF vectorization** layer.
- Trained a statistical **Naive Bayes model** to evaluate and execute high-speed, real-time message domain routing.
- Deployed a web-based **Streamlit dashboard** to visualize live classification maps and model confidence scores.

Visualizing Floor Plans Using Virtual Reality

June 2022 – June 2023

Technologies: Python, OpenCV, Blender, PyQt5

- Developed an **OpenCV image pipeline** using adaptive thresholding to parse structural layouts from 2D blueprints.
- Engineered **automated component detection** using ORB keypoint extraction to isolate doors and windows without labels.

Technical Skills

Languages: JavaScript, Python, C/C++, Java, Dart, SQL (PostgreSQL), HTML/CSS, R

Frameworks & Libraries: React Native, LightningJS, Flutter, Node.js, Express.js, Jest, Cypress, pandas, NumPy, Matplotlib

Developer Tools: Git, Docker, Terraform, Ansible, AWS (EC2, S3), VS Code, Jupyter

Operating Systems: Linux, Windows, macOS